

HANDWRITTEN DIGIT RECOGNITION USING KNN,SVM,CNN

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1. Aim of the Project

The aim of this project is to implement and compare different machine learning and deep learning algorithms-

K-Nearest Neighbors (KNN), **Support Vector Machine (SVM)**, and **Convolutional Neural Network (CNN)** for handwritten digit recognition using the MNIST dataset. The performance of each model is evaluated based on accuracy and confusion matrix analysis.

2. Introduction

Handwritten digit recognition is a fundamental problem in the field of pattern recognition and machine learning. It has practical applications in postal systems, bank cheque processing, document digitization, and optical character recognition (OCR).

In this project, both traditional machine learning algorithms (KNN and SVM) and a deep learning model (CNN) are implemented to understand their effectiveness and performance differences when applied to image-based data.

3. Dataset Description

The **MNIST dataset** is used for this project. It consists of grayscale images of handwritten digits from **0 to 9**.

- Image size: **28 × 28 pixels**
- Training samples: **56,000**
- Testing samples: **14,000**
- Total classes: **10 (digits 0–9)**

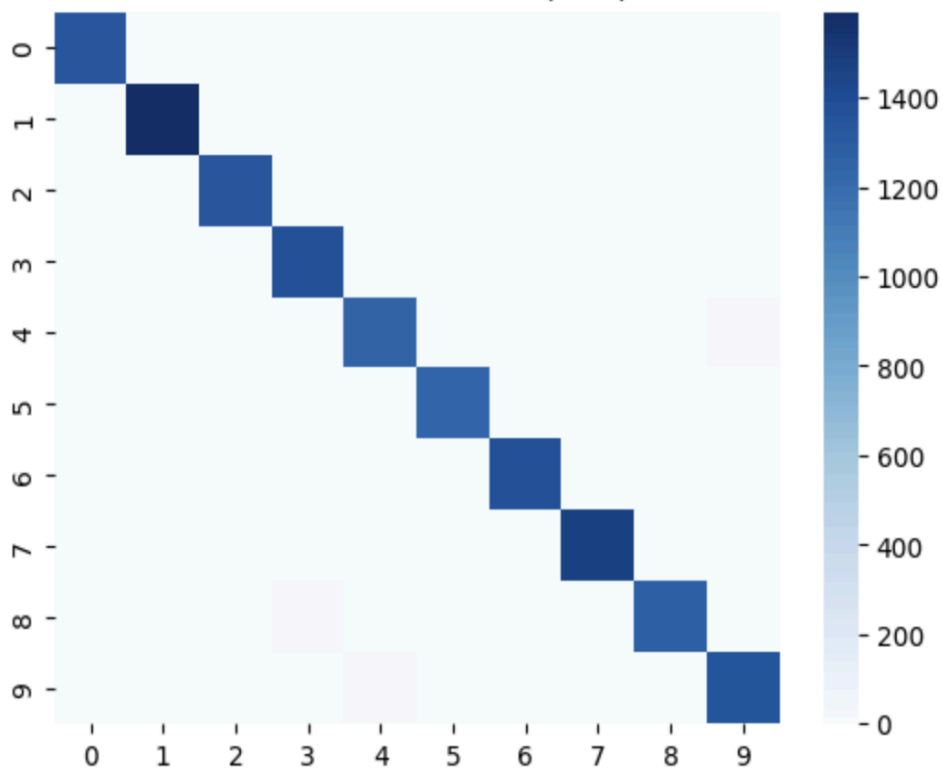
For KNN and SVM, each image is flattened into a vector of **784 features**.

For CNN, images are reshaped into **28 × 28 × 1** format to preserve spatial information.

4. Methodology

4.1 Data Preprocessing

- Pixel values are normalized by dividing by 255.



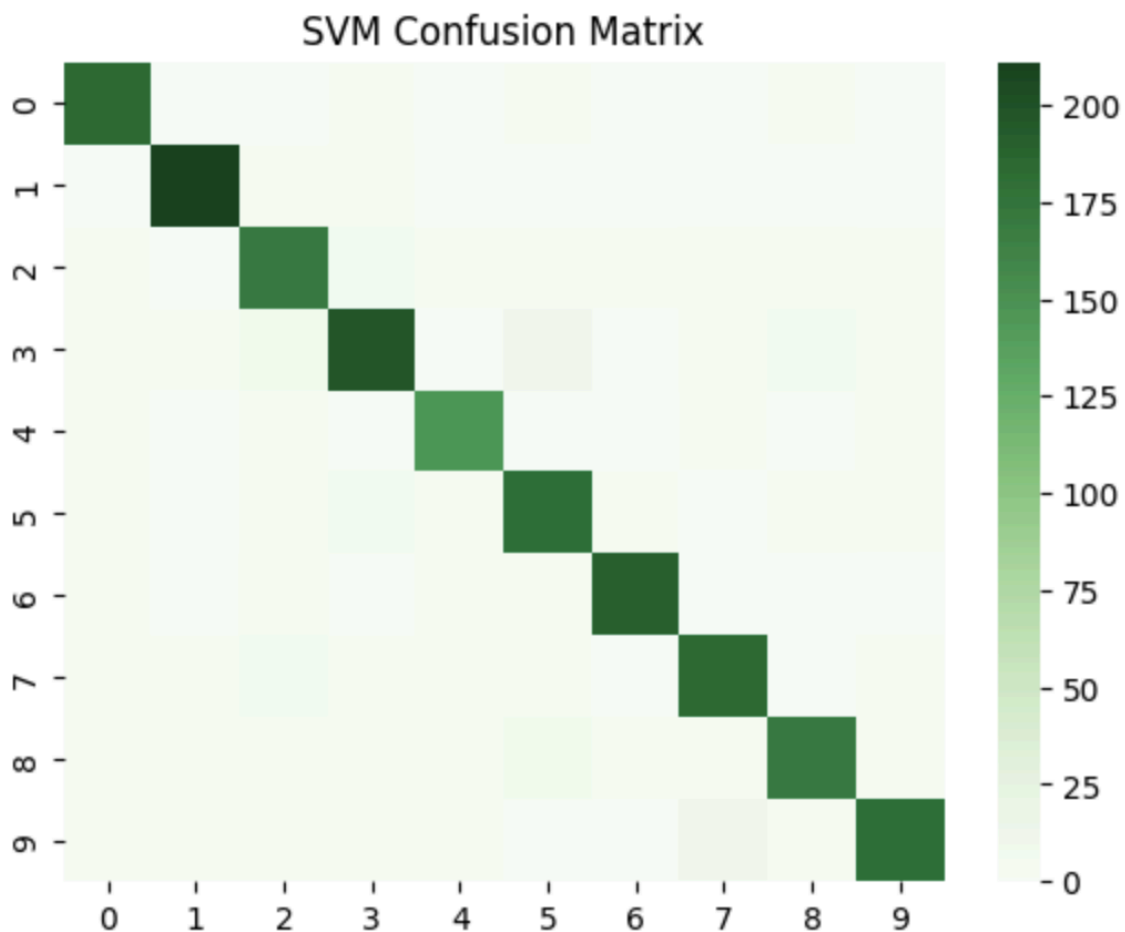
4.3 Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that finds an optimal hyperplane to separate classes.

- Kernel used: **Linear kernel**
- Regularization parameter: **C = 1**

SVM performed well but showed comparatively lower accuracy due to the high dimensionality of image data and lack of feature extraction.

SVM Linear Accuracy: 0.9075



4.4 Convolutional Neural Network (CNN)

CNN is a deep learning model specifically designed for image data. It automatically learns spatial features using convolution operations.

The CNN architecture includes:

- Convolutional layers with ReLU activation
- Max-pooling layers
- Fully connected (Dense) layers
- Softmax output layer for classification

CNN achieved the highest accuracy because it preserves spatial information and learns hierarchical features.

CNN Accuracy: 0.9829999804496765

5. Experimental Results

5.1 Accuracy Comparison

Model	Accuracy
KNN (k=3)	97.12%
SVM (Linear)	90.75%
CNN	98.30%

5.2 Confusion Matrix Analysis

- **KNN** shows strong diagonal dominance, indicating correct classification for most digits.
 - **SVM** has more misclassifications, especially for visually similar digits.
 - **CNN** produces the cleanest confusion matrix with minimal errors.
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6. Observations

- KNN performs well but is computationally expensive for large datasets.
- SVM struggles with image data without advanced feature extraction.
- CNN outperforms traditional methods due to its ability to learn spatial features.

- Deep learning models are more suitable for image recognition tasks.
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7. Conclusion

This project demonstrates that while traditional machine learning algorithms like KNN and SVM can achieve good performance on handwritten digit recognition, **Convolutional Neural Networks provide the best results**. CNN's ability to automatically extract relevant features makes it highly effective for image-based applications. Hence, CNN is the preferred model for handwritten digit recognition tasks.

8. Future Scope

- Use deeper CNN architectures for improved accuracy
 - Apply data augmentation techniques
 - Extend the system to recognize handwritten characters or words
 - Deploy the model as a web or mobile application
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9. References

1. MNIST Dataset – Yann LeCun
2. Scikit-Learn Documentation
3. TensorFlow & Keras Documentation
4. Machine Learning by Tom Mitchell